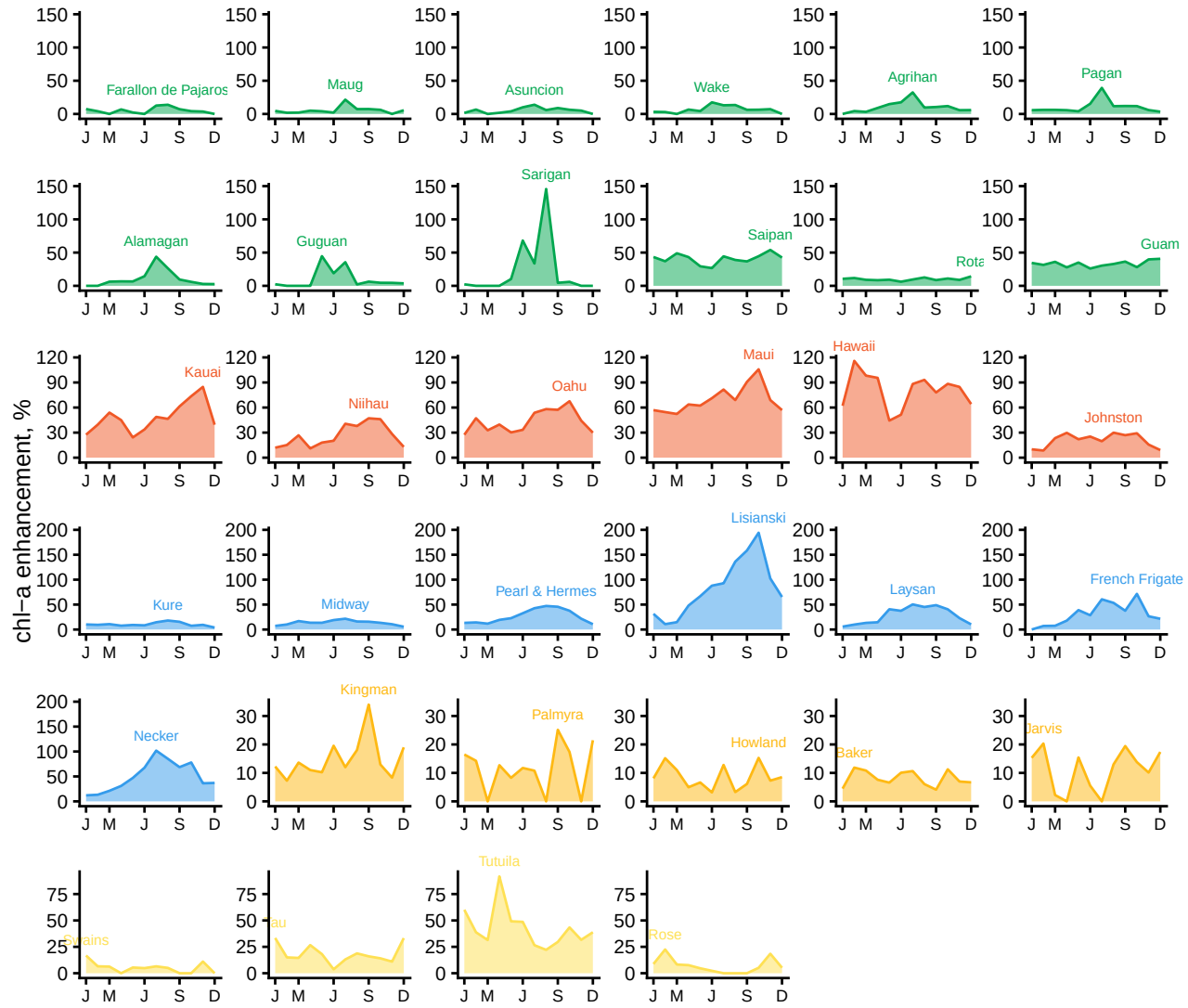
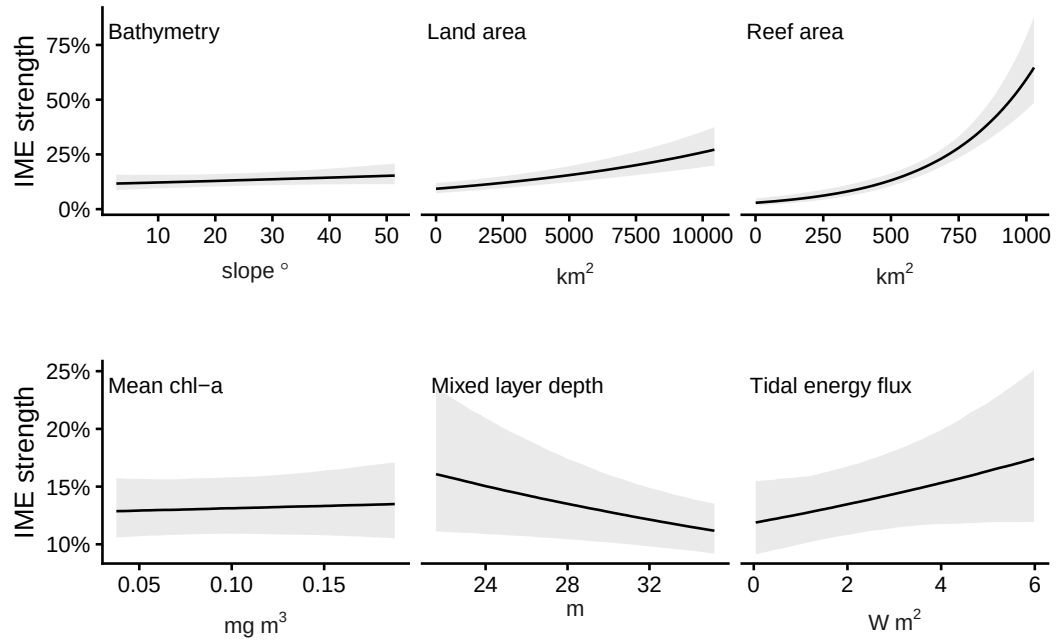


## Supplementary Figures

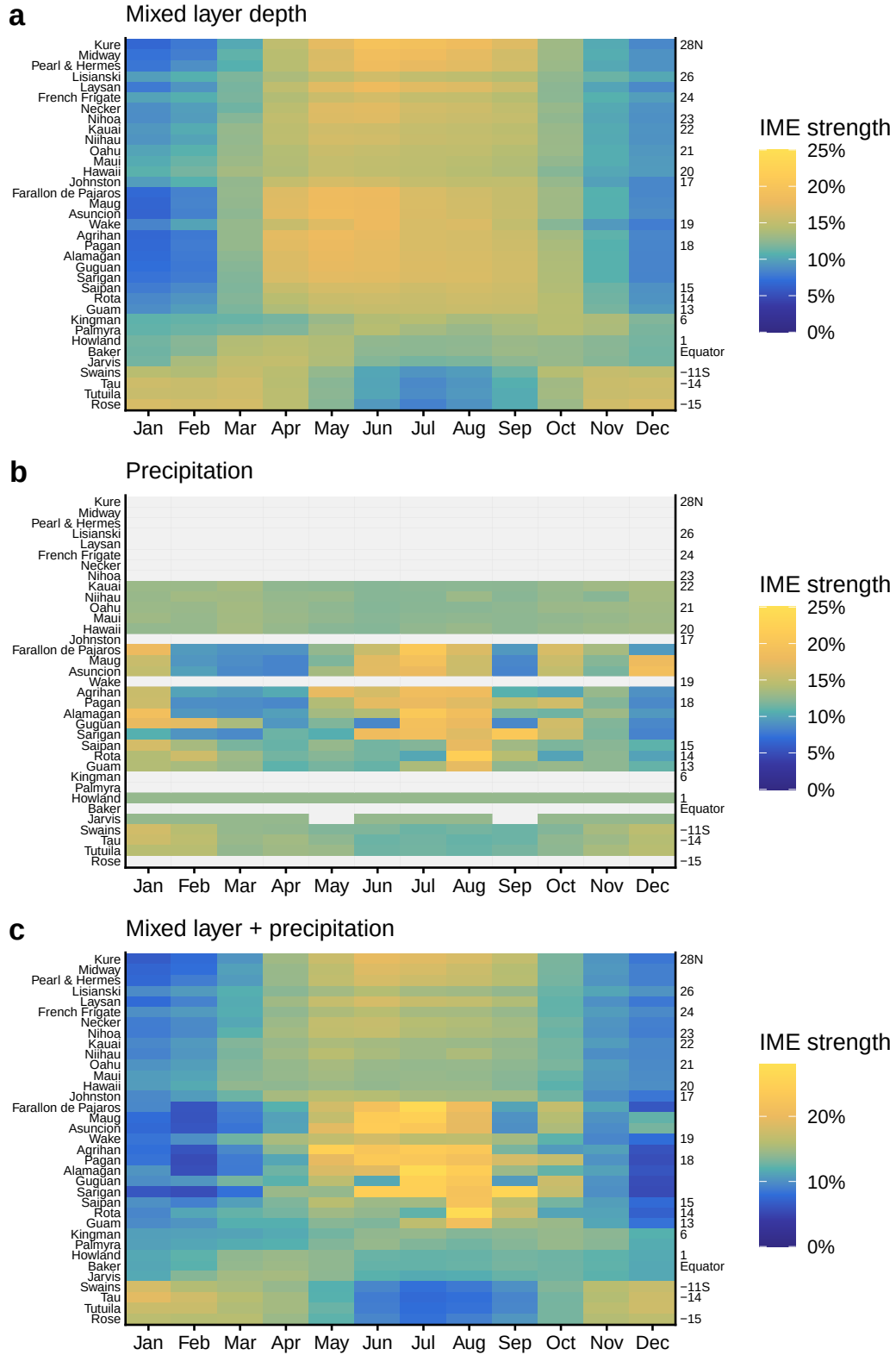
**Fig. S1** | *tbd* Map of 35 islands surveyed by CREP across the tropical Pacific.



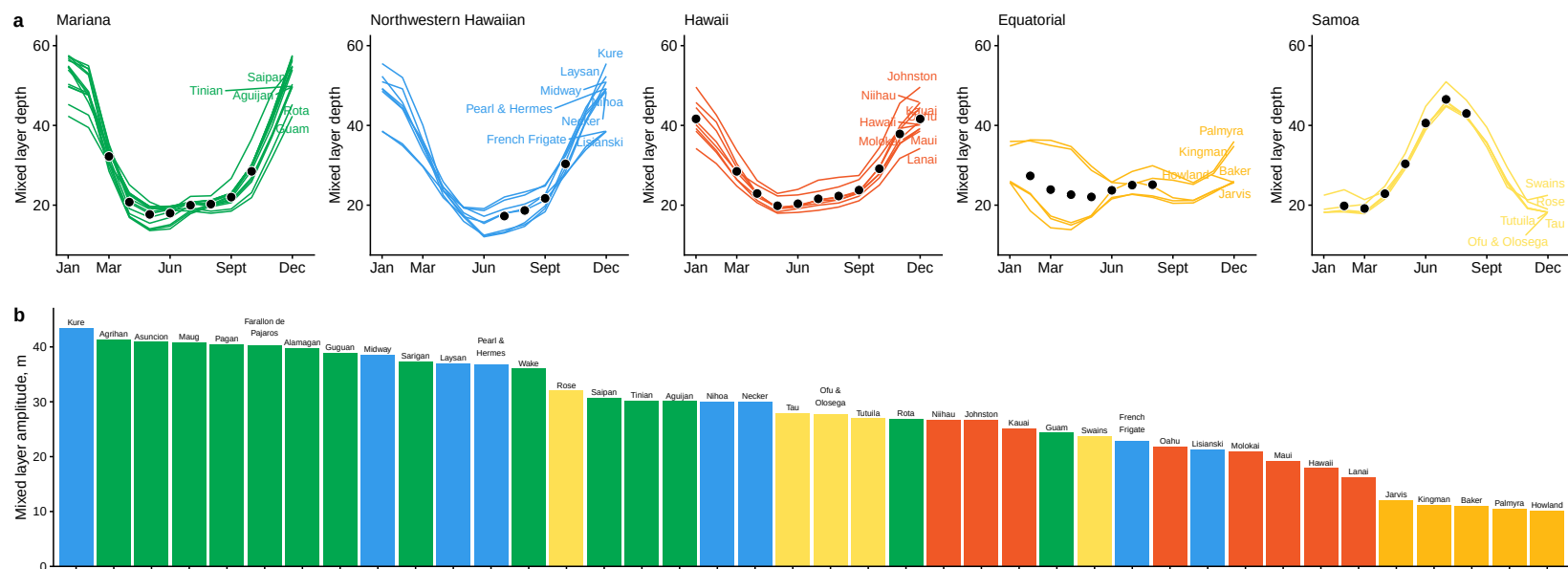
**Fig. S2** | Seasonal variation in IME strength at each island. Lines are mean-monthly chl-a enhancement from January to December, ordered and coloured by island group.



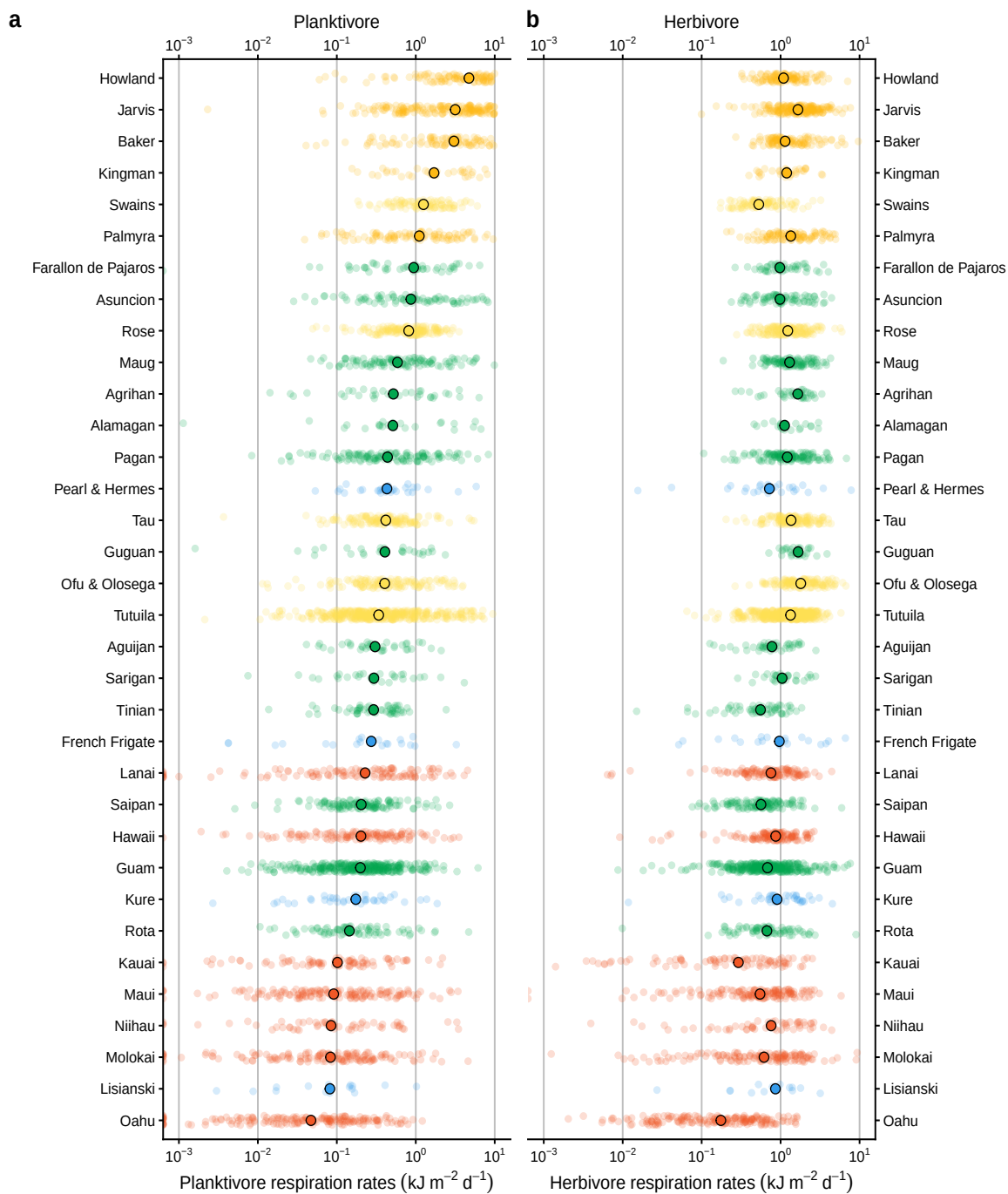
**Fig. S3** | Spatial effects on chl-a enhancement for geomorphic features (top) and average oceanographic conditions (bottom) Lines are median posterior predictions ( $\pm 95\%$  certainty intervals). Note different y-axis values between static and dynamic covariates.



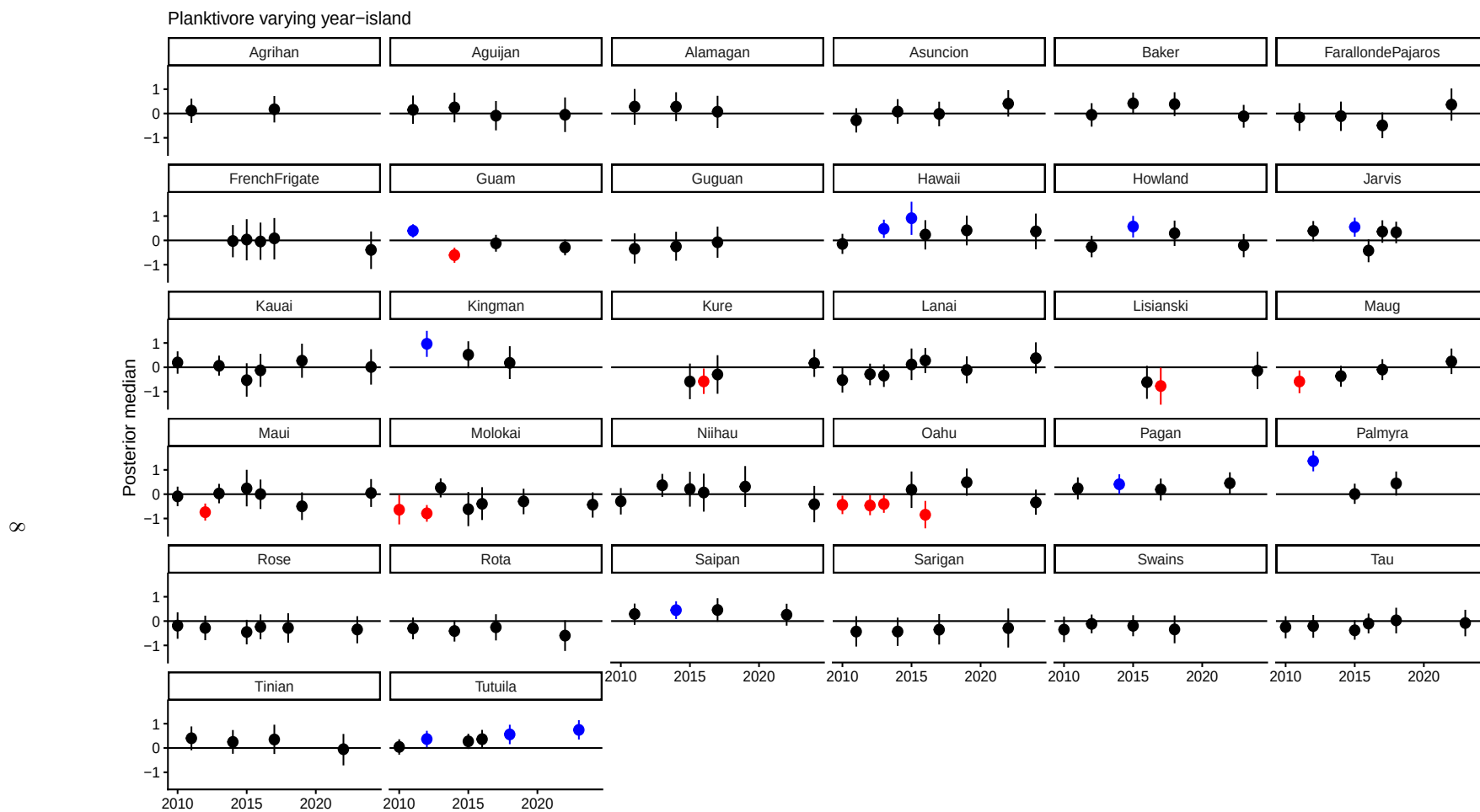
**Fig. S4** | Heatmap of seasonal effects on chl-a enhancement for a mixed layer depth, b precipitation and c both covariates combined. Islands with less than 1 mm monthly rainfall are shown in grey. All values are median posterior predictions excluding effects of other covariates.



**Fig. S5** | Seasonal variability in the mixed layer depth at each island. In a, lines are predicted mixed layer depth at each island from a General Additive Model with smoothers for time and month (cyclic), overlaid with points indicating timing of coral reef fish surveys.

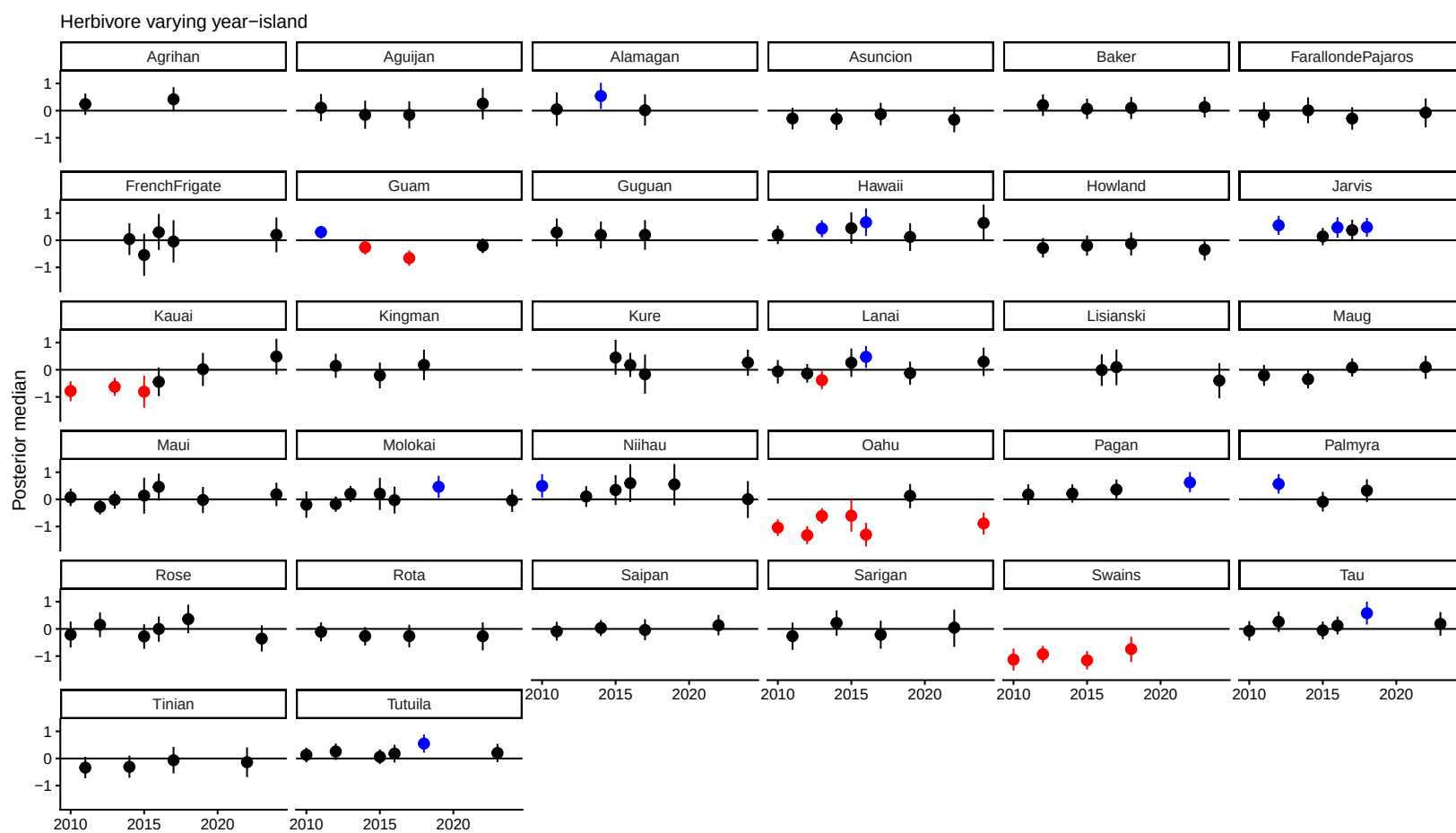


**Fig. S6** | Respiration rate of reef-associated planktivores on coral reefs at 34 Pacific islands and atolls. Each point is the median respiration at each island, with jitter showing unique sites ( $n = 3,622$ ).

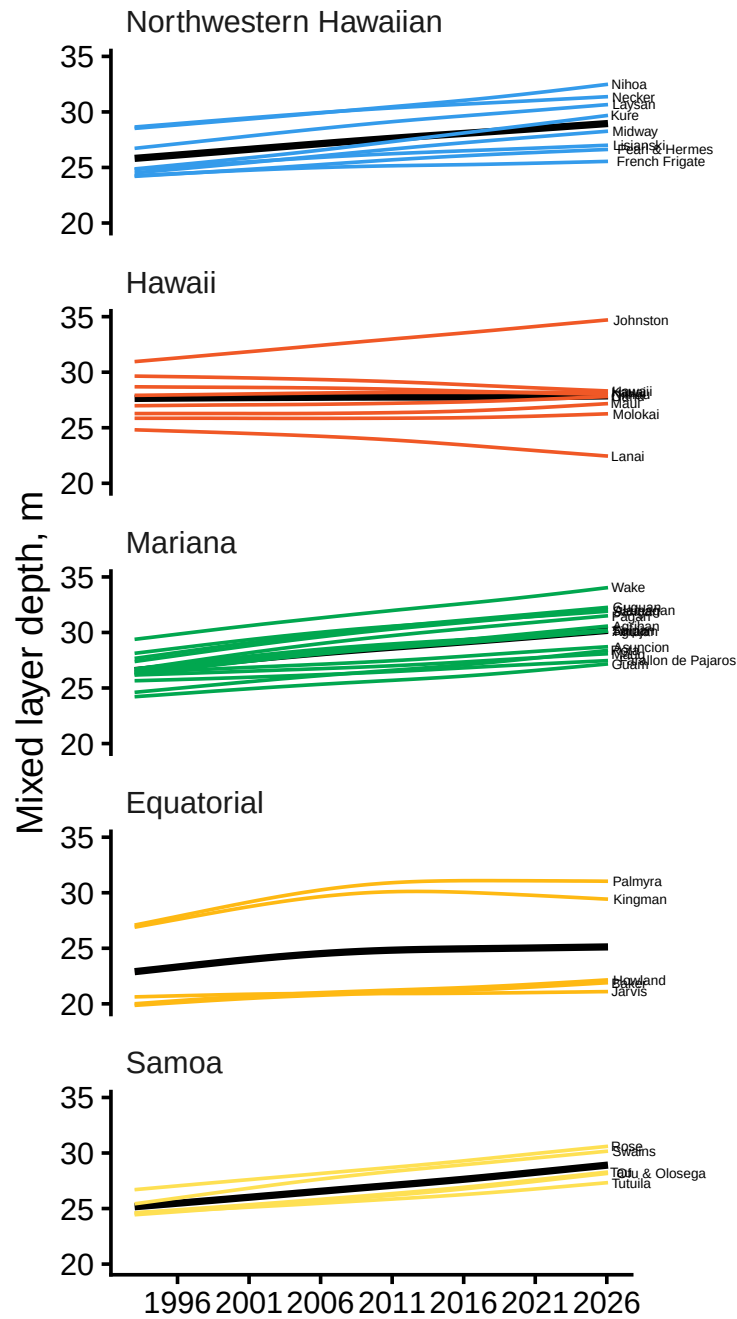


**Fig. S7** | Effect of survey year on planktivore respiration rate. Points are posterior medians with 95% certainty intervals, coloured when the posterior did not overlap zero (red = negative, blue = positive).





**Fig. S8** | Effect of survey year on herbivore respiration rate. Points are posterior medians with 95% certainty intervals, coloured when the posterior did not overlap zero (red = negative, blue = positive).



**Fig. S9** | Island-level temporal trends in the mixed layer depth layer from 1997-2026. Lines are mean predicted mixed layer from a generalised additive model, for each region (black with 95% CI filled by region), overlaid with mean island predictions (coloured lines).